

GE Healthcare

Optima MR450w
Field Replaceable Units (FRU)

Technical Publication

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GE Medical Systems, a General Electric Company, going to market as GE Healthcare
3000 N. Grandview Boulevard
Waukesha, Wisconsin 53188
USA

www.gehealthcare.com



5670005
Rev 1

**COVER
BOTTOM
CROP LINE**

**COVER
BOTTOM
CROP LINE**

**5/16" RADIUS
CORNER TRIM
TYPICAL**



**COVER
LEFT
CROP LINE**

**COVER
FOLD
LINE** | **(2) CENTER STAPLES
FOR BINDING**

**COVER
RIGHT
CROP LINE**



**5/16" RADIUS
CORNER TRIM
TYPICAL**

SEND COMMENTS ABOUT THIS MANUAL TO:

GE MEDICAL SYSTEMS
MANAGER, SERVICE ENGINEERING
W-808
3200 N. GRANDVIEW BLVD.
WAUKESHA, WISCONSIN 53188
USA

This manual belongs to:

INTRODUCTION TO FIELD REPLACEABLE UNIT (FRU)

This direction is a guide to the FRU's.

Parts designated as a FRU are critical to maintain the safety and/or functionality of the system. FRU's are available on a Priority 1 basis from Service Parts Operation.

As a reference aid, the FRU's are listed under an upper-level part number where they are used. In most cases the upper-level part number is not a FRU. In some cases the parts listed under an upper-level assembly may have been nested in intermediate sub-assemblies which are not shown.

Where possible, FRU's which are part of Option Catalogs are shown under the area where they would be installed, rather than being listed in the Coils or Options Sections (SEE TABLE OF CONTENTS).

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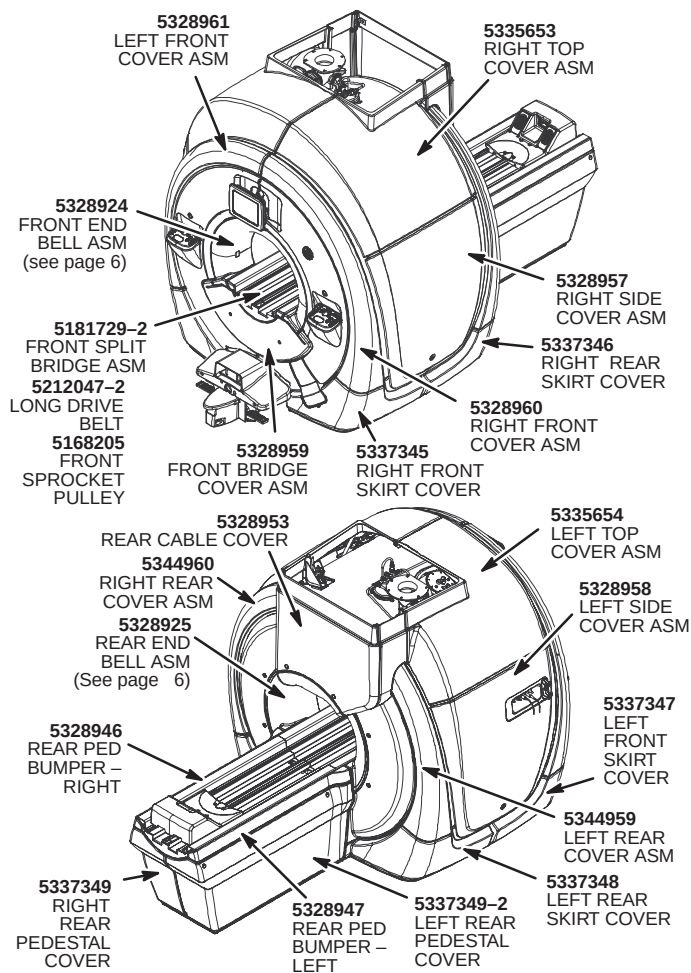
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46-255325P1	Security Zone Warning Sign
46-255814P1	Magnetic Warning Label, ~2.5 x ~1.5
46-301671P1	Warning Placard
2289812	Authorized Personnel Only Warning Sign

[illegible]

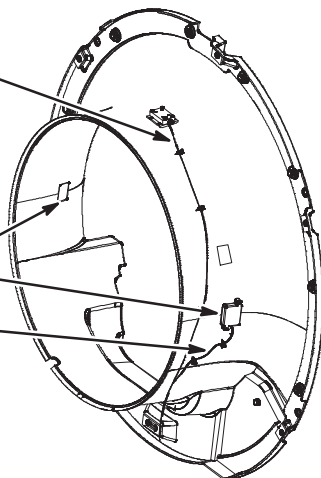
ENCLOSURE COMPONENTS**MAGNET ENCLOSURE**

END BELL COMPONENTS**5328924 FRONT END BELL**

5183293
MICROPHONE CABLE ASM

5344830
FRONT BORE LIGHT ASM

5344941
ENDBELL BORE LIGHT CABLE

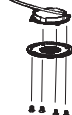
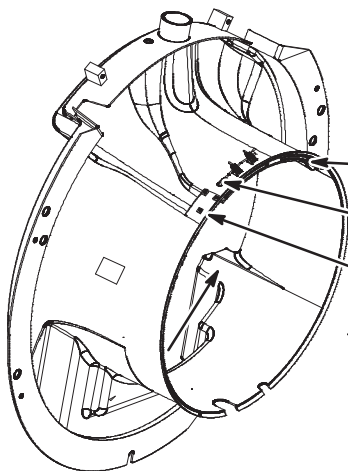
**5328925 REAR END BELL**

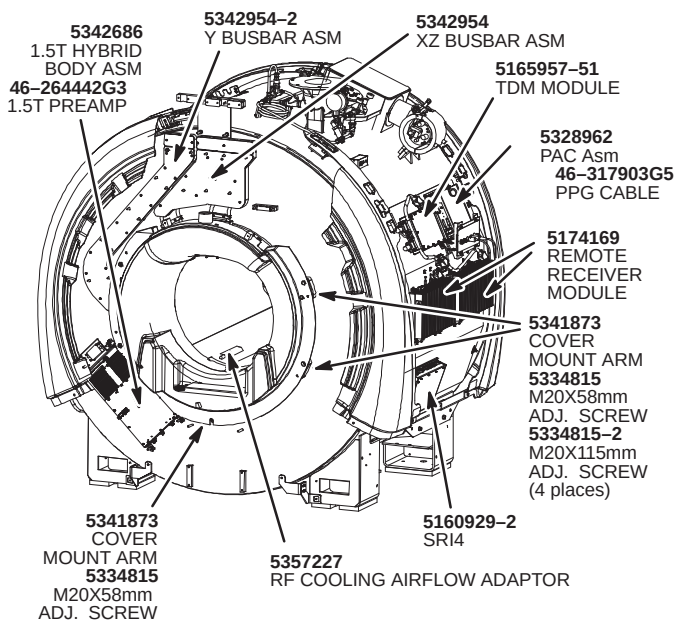
5339429
BORE LIGHT ASM – RIGHT

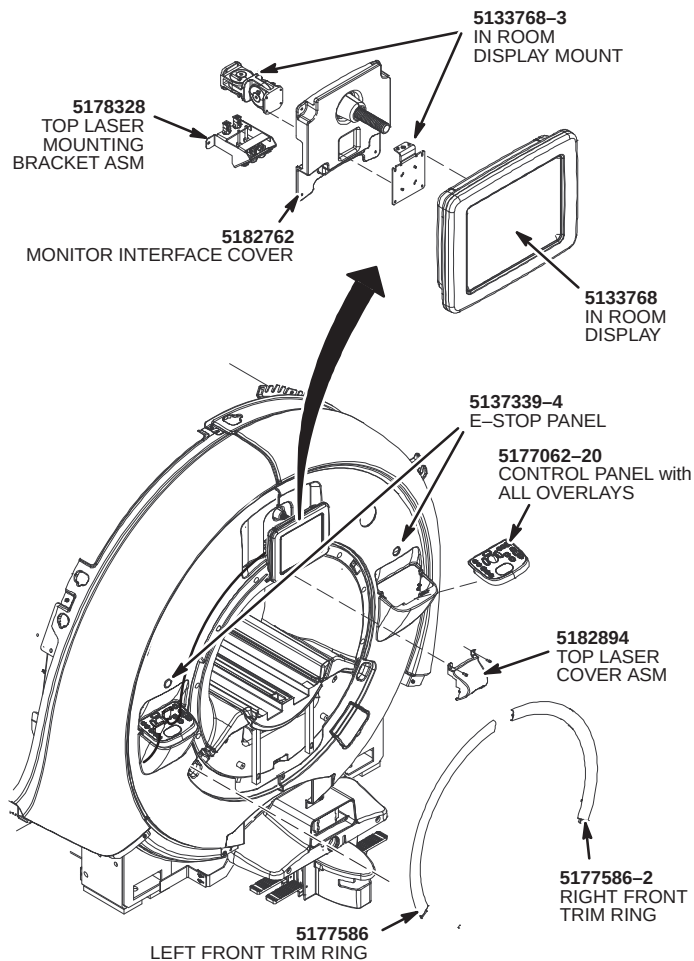
5344941
BORE LIGHT CABLE

5339427
BORE LIGHT ASM – LEFT

4602221773P2
SPEAKER

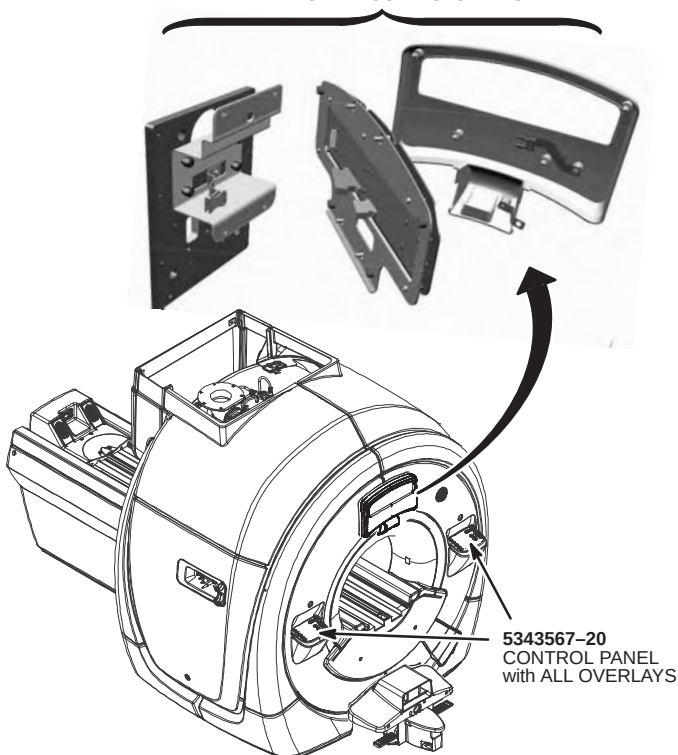


ENCLOSURE ELECTRONIC COMPONENTS**MAGNET FRU's****46-260852G3** Emergency Evacuation Release Tool**5366120** Stud Replacement Kit**CABLE FRU's****2128699** Run 606, Magnet to ERU/MRU**5183293** Microphone Cable, Run M3345**5137339-6** E-Stop Cable**5180907** Run 3350, MAG-SRI-J10 to MAG-BORE, Bore Temp,**SCAN ROOM FRU KIT's****5196918** MRU Assembly**2394952** Magnet Monitor 3**2219341** Ruthenium Oxide Sensor Amplifier**5155291-2** Magnet Service Platform**MAGNET ENCLOSURE**

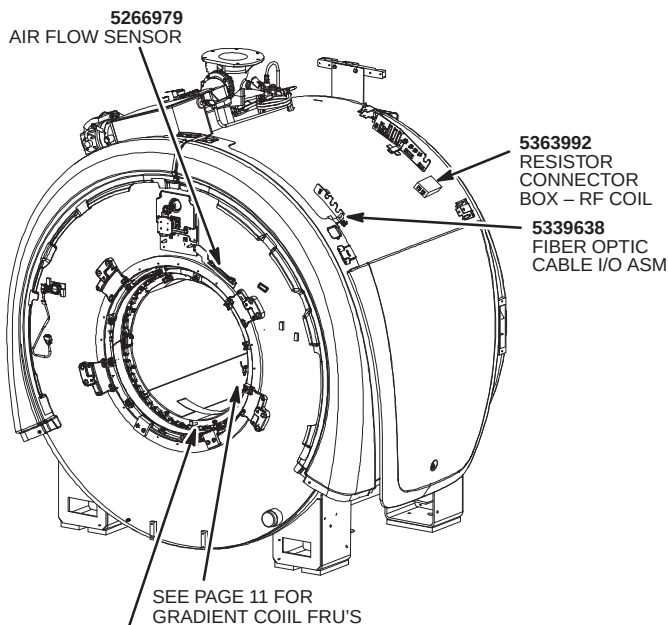
ENCLOSURE COMPONENTS for IN-ROOM DISPLAY (IRD)

ENCLOSURE COMPONENTS for LED DISPLAY

5351731
LED DISPLAY COLLECTOR FRU



MAGNET ENCLOSURE

ENCLOSURE AND SHIM COMPONENTS

5329393
XRMW SHIM TRAY ASM

5342951
FIELD PASSIVE SHIM FRU KIT contains:

5343857	Shim, Silicon Stl 2.1mm thick, M4 Thru (40)
5343857-2	Shim, Silicon Stl 4.9mm thick, 4.5 Dia (40)
5343857-3	Shim, Silicon Stl .35mm thick, 4.5 Dia (250)
5343857-4	Shim, Silicon Stl Half .35mm thick, 4.5 Dia (120)
5343857-5	Shim, Silicon Stl 3.5mm thick, M4 Thru (40)
5343857-6	Shim, Silicon Stl 3.5mm thick, 4.5 Dia (50)
5343857-7	Shim, Silicon Stl 4.9mm thick, M4 Thru (50)
5385706-5	M4 x 10mm Brass flat Head Screw (75)
5212343	Passive Shim Square Nut (150)
5214479	Passive Shim Screw (100)

MR450w 1.5T GRADIENT AND RF BODY COIL**2208999-15**

1.5T RF BODY COIL Includes
BLUE & BLACK FRU CASE
AND SKID

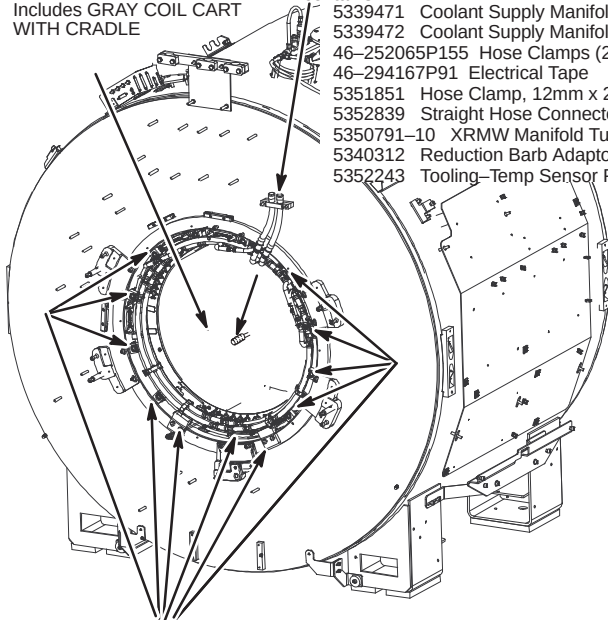
5342791

XRMW GRADIENT COIL
Includes GRAY COIL CART
WITH CRADLE

5342778

COOLANT MANIFOLD COLLECTOR FRU
contains:

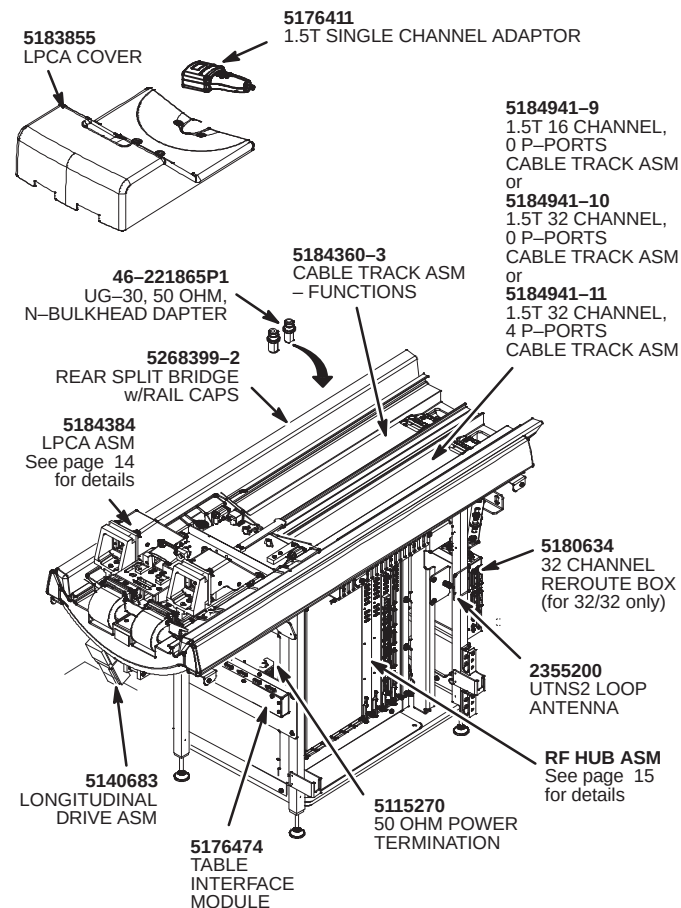
- 5339471 Coolant Supply Manifold – BLU
- 5339472 Coolant Supply Manifold – BLK
- 46-252065P155 Hose Clamps (25)
- 46-294167P91 Electrical Tape
- 5351851 Hose Clamp, 12mm x 20-32
- 5352839 Straight Hose Connector, 12mm
- 5350791-10 XRMW Manifold Tube
- 5340312 Reduction Barb Adaptor
- 5352243 Tooling-Temp Sensor Protector

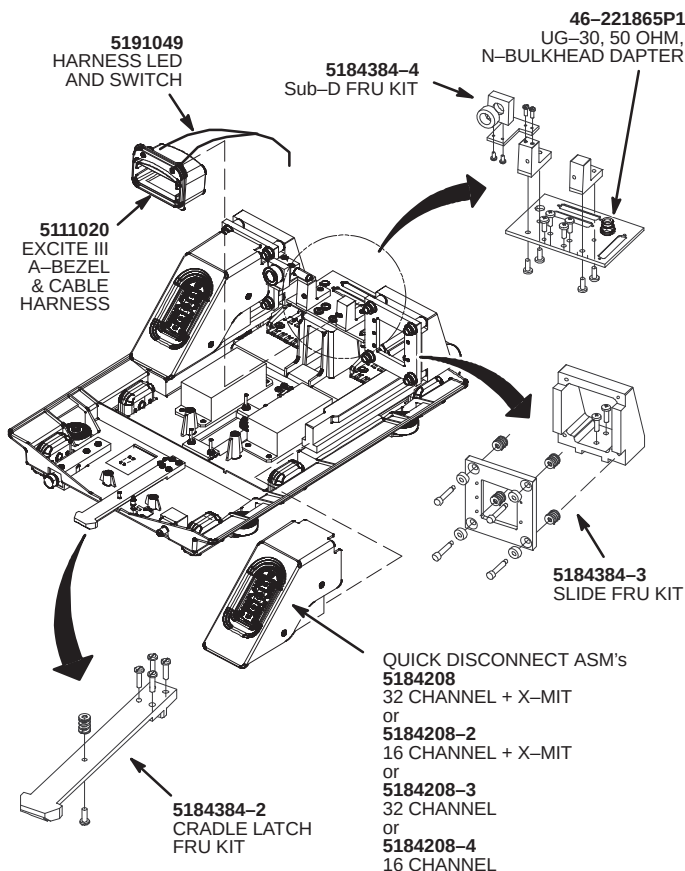
**5342776**

XRMW GRADIENT MOUNTING FRU KIT contains:
(For mounting front and rear of coil)

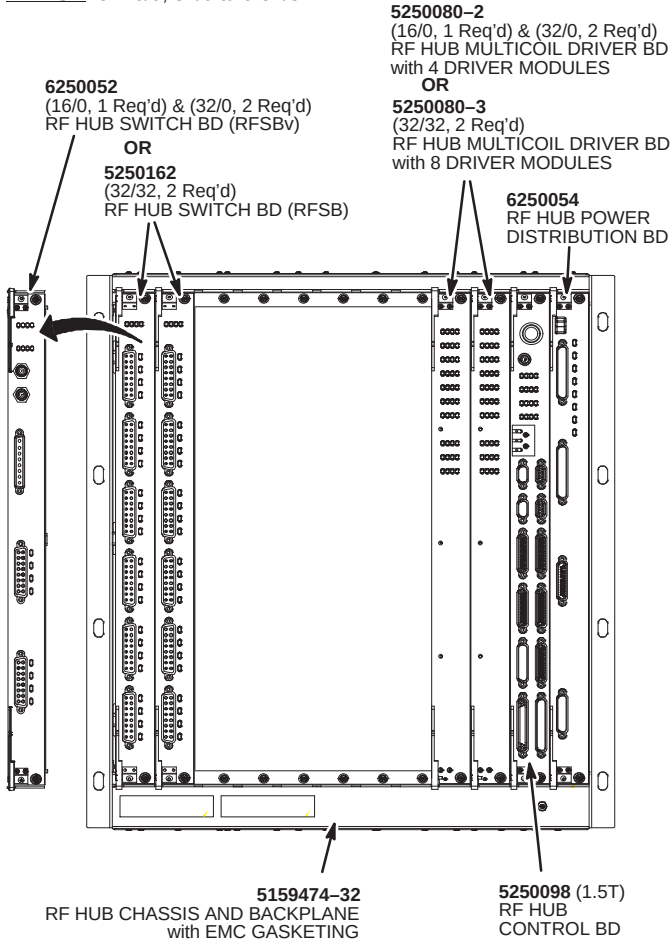
- 5352690 Mounting Wedge (20)
- 5352691 Mounting Stop Block (4)
- 5352696 Mounting Pad (2)
- 5352697 Mounting Block Shim (16)
- 5363536-20 M10 x 30mm Hex Head Cap Screw (8)
- 5110638-65 M10 X 35mm Brass Hex Head CS (20)
- 5146169 M10 x 24mm Washer (28)
- 46-294151P11 Locktite 271

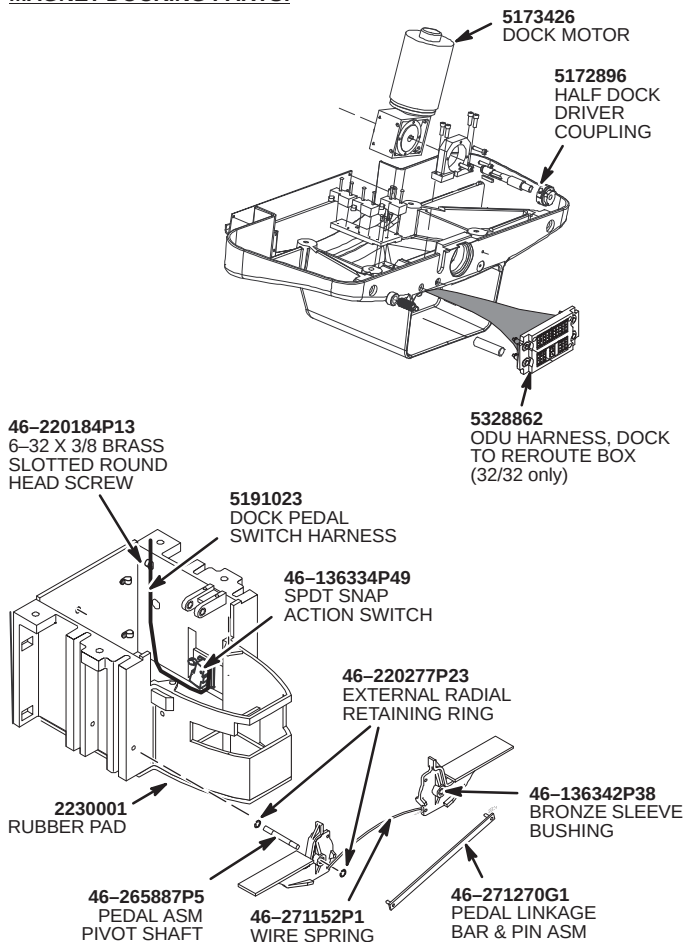
[illegible]

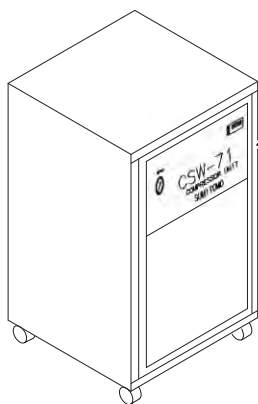
REAR PEDESTAL

LPCA

RF HUB for 16/0, 32/0 and 32/32



MAGNET DOCKING PARTS:

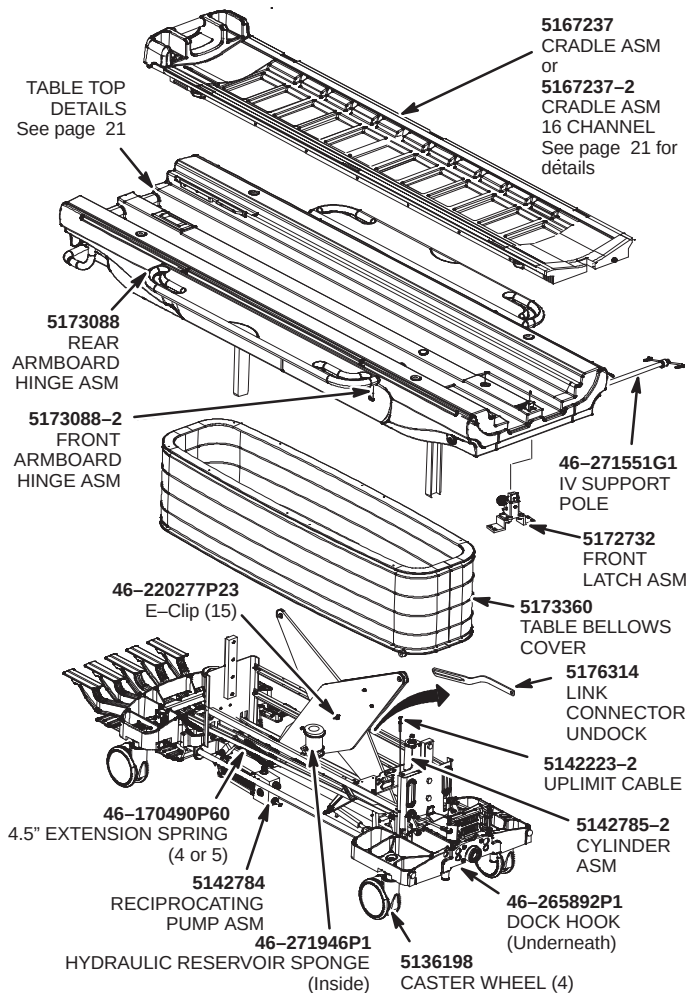
SHIELD COOLER & COLD HEAD

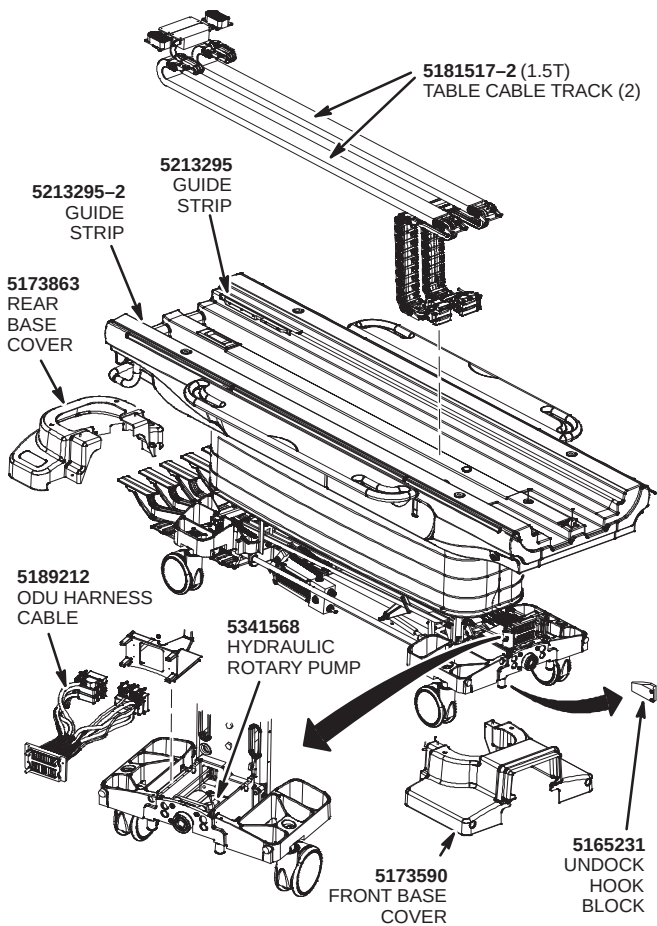
5314887
Sumitomo F-50
Shield Cooler Compressor

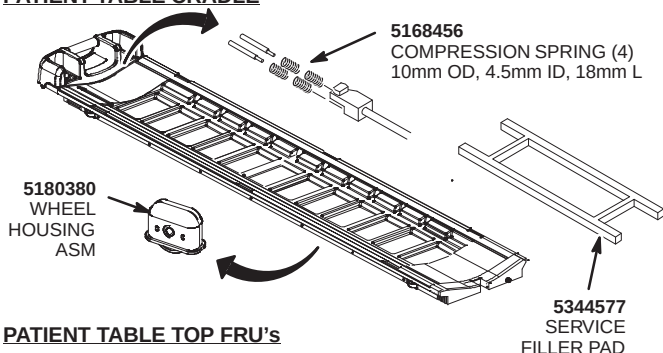
SHIELD COOLER FRU's Not Shown:

- 2154502-2** Run 621, Supply Gas Line, 65.6 ft or 20M long
- 2154505-2** Run 622, Return Gas Line, 65.6 ft or 20M long
- 5271637** Return Gas Line Adaptor, F-50 Compressor
- 5265687** Supply Gas Line Adaptor, 180 Degrees, F-50 Compressor
- 2155316-2** Run623, Compressor to Pen Panel Cable, 49.21 ft or 15M long
- 2218192** Run624, Cold Head Control Cable
- 2214062** Run 832, LCC Sleeve and Cold Head Cable
- 2259837** LCC Cold Head Control Cable Asm
- 2200835** Power Cable for Sumitomo Shield Cooler Compressor
- 5335002** Flow Control Adapter, used with F-50 Compressor only

[illegible]

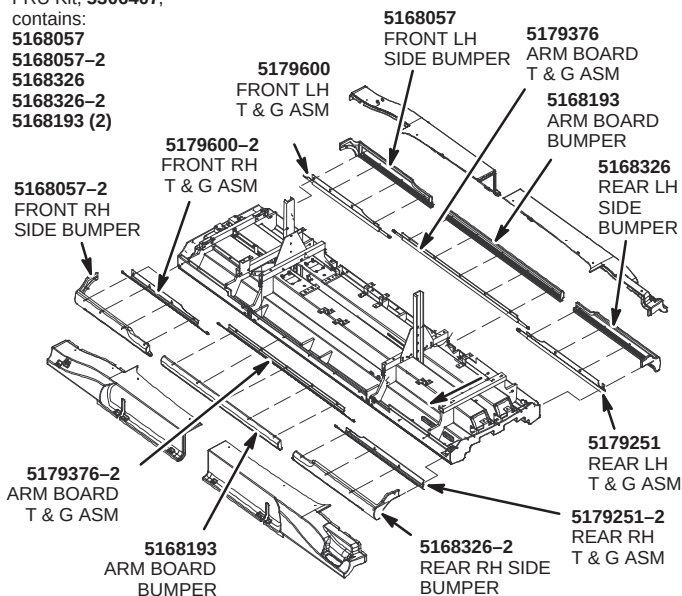
PATIENT TABLE**PATIENT TABLE**

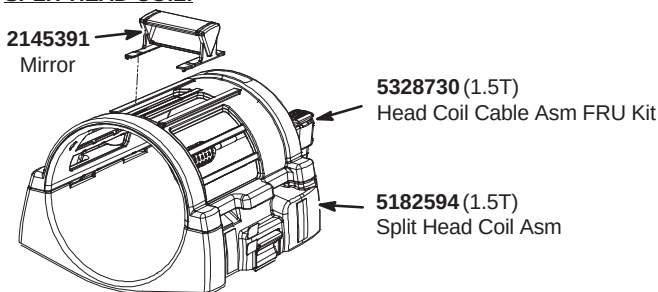
PATIENT TABLE

PATIENT TABLE CRADLE**PATIENT TABLE TOP FRU's**

The Table Bumpers
FRU Kit, 5306407,
contains:

5168057
5168057-2
5168326
5168326-2
5168193 (2)

**PATIENT TABLE**

SPLIT HEAD COIL:**Note: The following parts are not shown:**

5328732 (1.5T) QHTR Circuit Board Asm Kit

5267144 Head Coil Cushion – Flat

5271434 Head Coil Cushion – Head

2355094 Replacement Spring Contacts Kit

PATIENT HANDLING FRU KIT's:

46-306574P7 0.5 x 9 x 14.5 Gray Rectangular Pad, Nylon Covered

46-306574P8 1.0 x 9 x 14.5 Gray Rectangular Pad, Nylon Covered

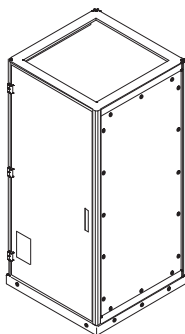
46-306574P9 0.5 x 9 x 36 Gray Rectangular Pad, Nylon Covered

46-306574P10 1.0 x 9 x 36 Gray Rectangular Pad, Nylon Covered

46-317758P1 Pneumatic Patient Alert System

5122500 ECG Lead Kit

HEC CABINET

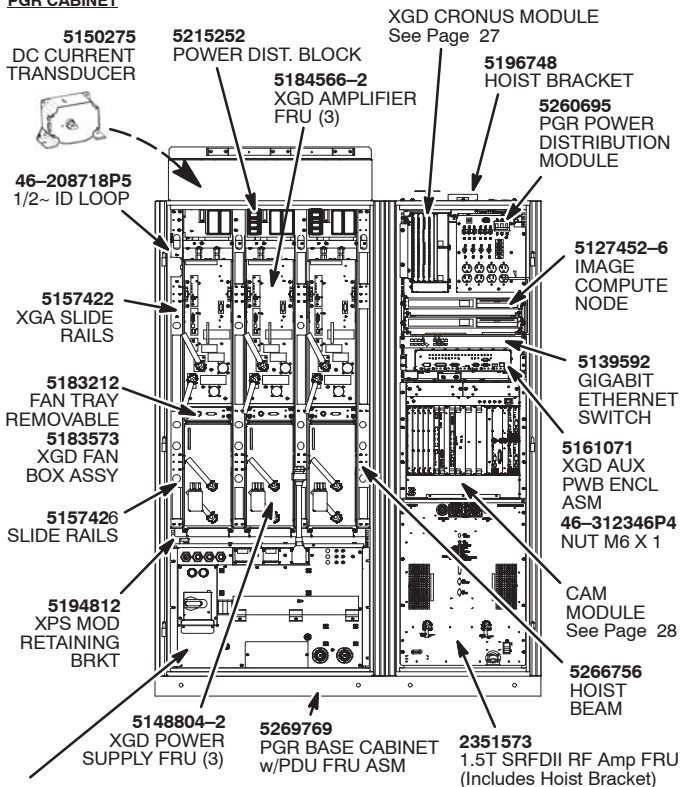


HEC (Heat Exchanger Cabinet) FRU KIT's

5264701	Deionization Filter Cartridges Kit
5156315	3hp Blower Kit
5156316	Cryogen Compressor Flow Meter Kit
5156318	Mixing Valve Kit for Gradient Coil and PGR Cooling Loops
5156319	Power Box Kit
5156320	Mixing Valve Kit for Chilled Air Blower
5156321	5hp Pump Kit
5262029	Equipment Room Hoist Support Tube
5266827	Fuse Kit
5267643	Liquid Pressure Sensor Kit
5267644	Level Switch Kit
5267648	Pressure Difference Sensor Board Kit
5267649	Resistivity Sensor Kit
5268665	Liquid Temperature Sensor Kit
5269683	Coolant Removal Kit
5269685	Cryogen Compressor Filter Kit
5269689	Air Temperature Sensor Kit
5269691	Signal Box Kit

HEC COOLING SYSTEM FRU's

5267596	Liquid Cooling Valve Kit (Contains ball valves and check valves)
5174313-4	Gradient Coil Coolant, 4 Gallons – non-copper corrosive, algae resistant

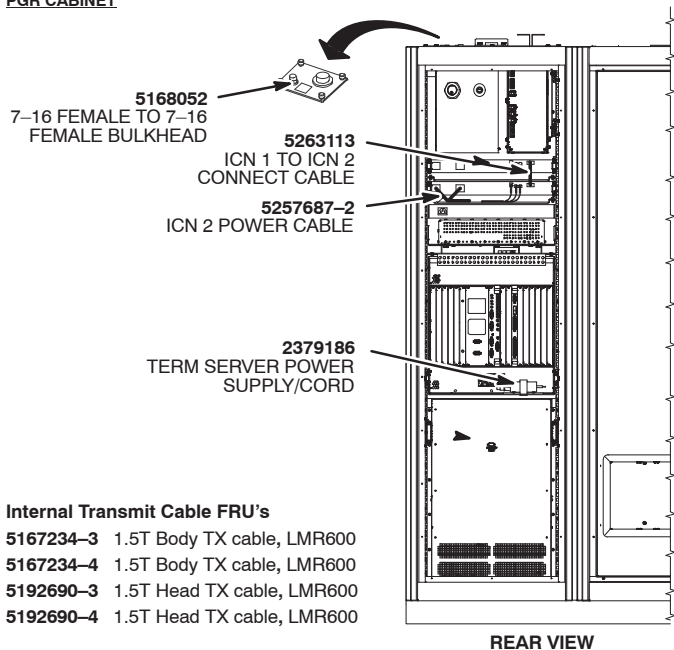
PGR CABINET**XGD PDU FRU'S (Not Shown)**

- 2301200-3** TVSS Module
- 5140621-2** XGD PDU Control Board Asm
- 5140621-3** 3 Pole Power Contactor for E-Stop
- 5140621-4** 48 Volt 1000W Power Supply
- 5140621-5** Fan with Tach Output

OTHER CABINET FRU'S (Not Shown)

- 5266125** Fluid Sensor Kit
- 46-306565P3** Aluminum Mesh Air filter, 16 x 10
- 2109873-25** M16 X 16mm L Pan Head Screw

PGR CABINET

PGR CABINET**Internal Transmit Cable FRU's**

- 5167234-3** 1.5T Body TX cable, LMR600
- 5167234-4** 1.5T Body TX cable, LMR600
- 5192690-3** 1.5T Head TX cable, LMR600
- 5192690-4** 1.5T Head TX cable, LMR600

5304065, Internal Signal Cable FRU Kit contains:

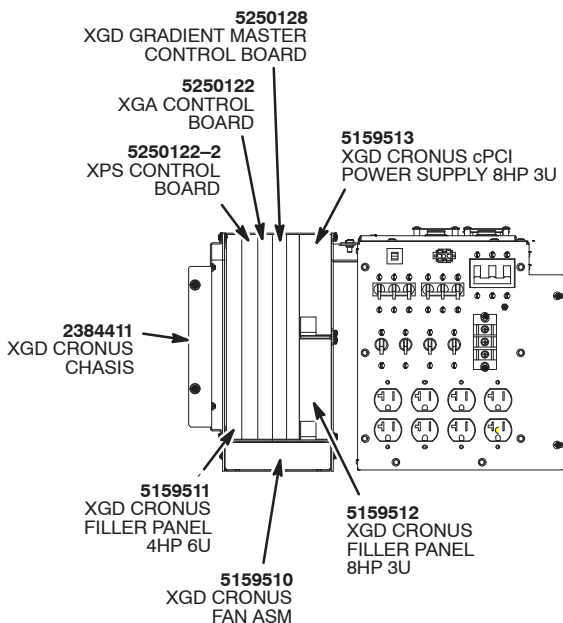
5178931	5165870	5165970	5176566	5182238	5182238-2
5182238-3	5182238-4	5184303	5257682	5257683	5257683-2
5257684	5257685	5257685-2	5257685-3	5257685-4	5257685-5
5257685-6	5257685-10	5174739	5167344	5167005	5257680
5257681	5166354	5166222-13	5166222-18	5257685-11	5257685-12
5257685-13	5176990	5176262	5176262-2	5176262-3	

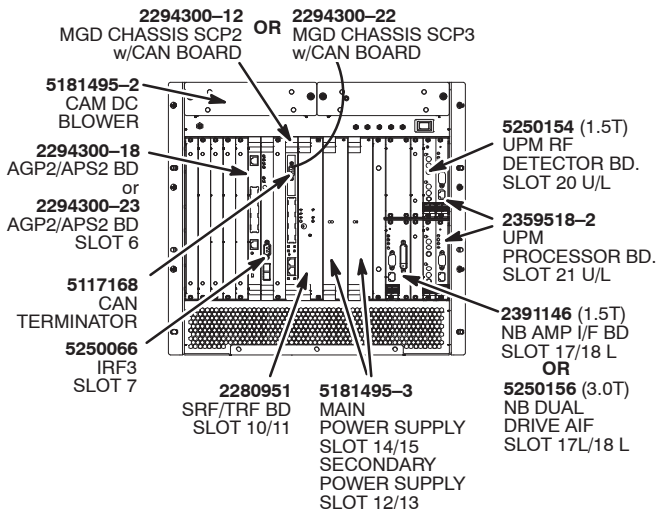
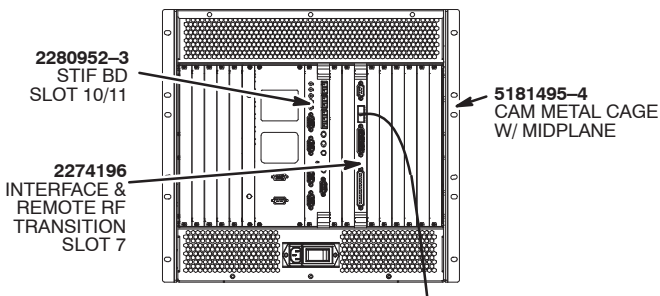
5304386, Internal Fiber Optic Cable FRU Kit contains:

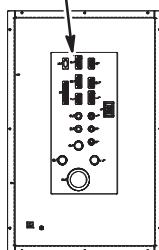
5182888	5182317	5182317-2	5182317-3	5182317-4	5182317-5
5182317-6	5257686				

5304635, Internal Power Cable FRU Kit contains:

5173399	5164904	5165800	5172391	5174242	5174242-2
5183977					

CRONUS MODULE

CAM MODULE (FRONT)**CAM MODULE (REAR)**

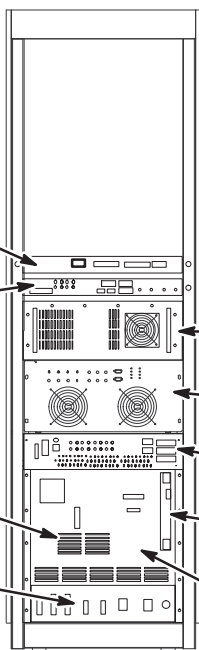
PEN CABINET**5199042
TOP IO PANEL****Top of Pen Cabinet**

5167035
FIBER OPTIC
CAN ASM

5154662 (1.5T)
REFERENCE
CLOCK & NB
EXCITER MODULE

5263700
FAN ASM

5166815-2
POWER
DISTRIBUTION
UNIT

**FRONT VIEW**

2352972
SHIM SUPPLY
MODULE

5160446-2
DRIVER
MODULE FRU

5167111
PATIENT HANDLING
POWER SUPPLY 2

5251938
SRPS CONTROL
BOARD ASSEMBLY

5231801-3
SCAN ROOM POWER
SUPPLY 16 & 32 CH

5117168
CAN TERMINATOR
(J4 on back of SRPS)

INTERNAL CABLE FRU KIT'S**5308001-2, Power Cable FRU Kit contains:**

5167143	5167143-2	5172838	5155030	5172849	5171737
5154994					

5308001-3, RF Cable FRU Kit contains:

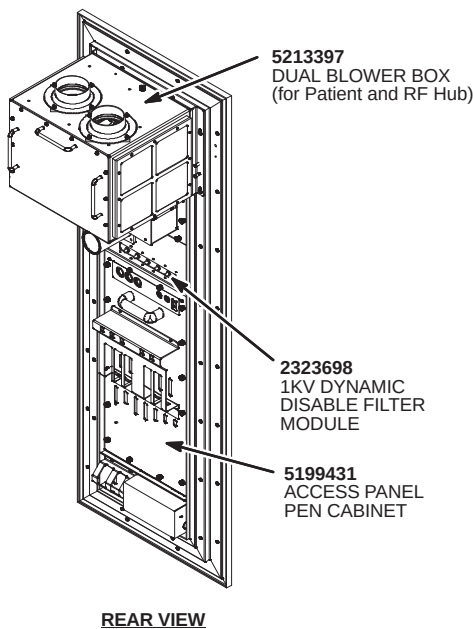
5181464	5166512-3	5166512-2	5172955-2	5172955	5166749
5166749-9	5172679-10	5172679-11	5172679		

5308001-4, Power Distribution Cable FRU Kit contains:

2322434	5168641	5168043	5168317	5167882	5166461
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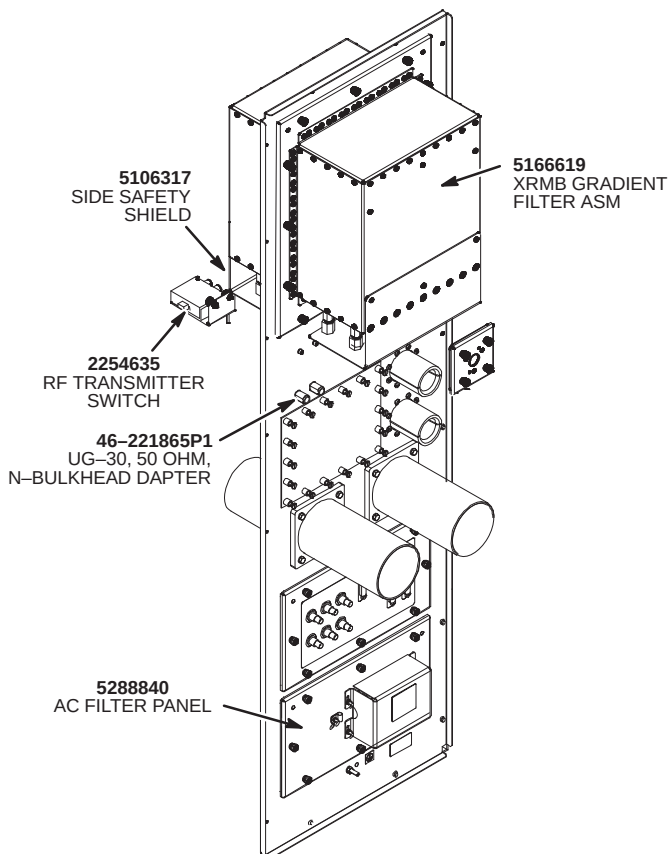
PENETRATION CABINET

PEN CABINET PENTRATION PANEL



INTERNAL CABLE FRU KIT'S (Continued from previous page)

5308001-5, Signal Cable FRU Kit contains:					
5165866	5166997	5169443	5172726	5166354-2	5171626
5170521	5172868	5167365	5171583	5172959	5169381
5168461	5168597	5172812	5172595-2	5169345	5175777
5172595					

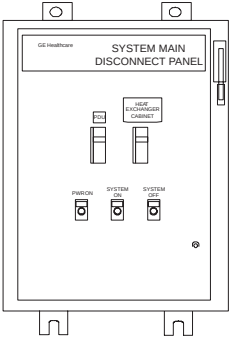
SPW (Secondary PEN Wall)

SPW

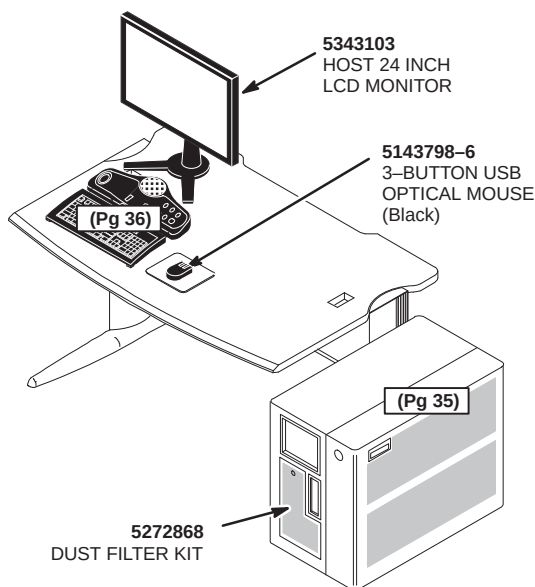
31

[illegible]

MAIN DISCONNECT PANEL (MDP)



- MDP FRU's Not Shown**
- 5161287-2** 1.5A, 380/480V Fuse
 - 5161287-3** 60A, 380/480V Fuse
 - 5161287-4** Green PB Lens
 - 5316807** PULS ML 15-121 Power Supply
 - 5316808** BVUPS 12v FRU Asm
 - 5340703-2** 1A 500V Fast Acting Fuse
 - 5323547** Power Sonic 12V 7.0AH Battery

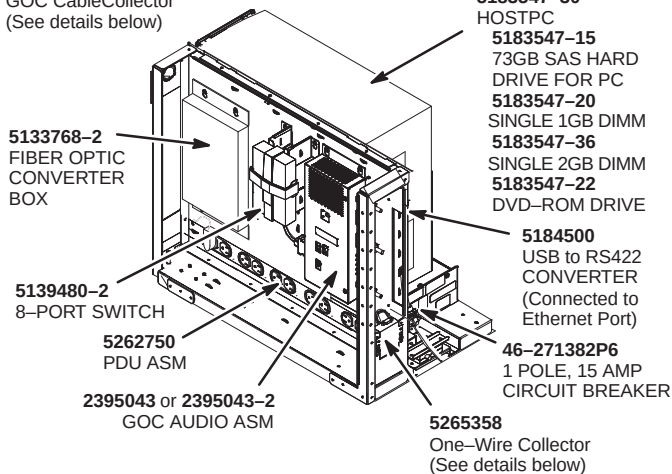
OPERATOR WORKSPACE

OPERATOR WORKSPACE FRU's Not Shown

5165713 DVD Media for Linux

GOC COMPUTER CABINET WITH HP8400;**5267342**

GOC CableCollector
(See details below)

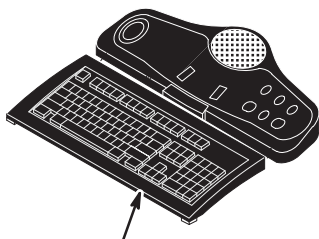


5267342 GOC Cable Collector FRU consists of;

- 5270774 _____ Host Power Cable
- 5270803 _____ Main LCD Monitor Power Cable
- 5263413 _____ Run E3046 Main Monitor DVI-D Cable
- 5264533 _____ Run E3047 SCIM-GOCOA Cable
- 5267826 _____ Audio Cable Asm
- 5270723 _____ GOCOA-PDU DC Power Cable
- 5267389 _____ Run E3045 Modem Cable
- 5272839 _____ Host Hub Ethernet Cable
- 5272839-2 _____ Hub-Front Port Ethernet Cable
- 5264521 _____ Run E3048 USB Extension Cable
- 5169105 _____ Run E3016 Y6 Cable
- 5265059 _____ Host to IRDC DVI-D Cable
- 5265784 _____ Host to IRDC USB Cable
- 5270246 _____ IRDC Power Cable

5265358 One-Wire Collector FRU consists of:

Humidity & Temperature iButton, iButton Holder, USB to RS422 Converter, USB and Sensor cables, Run E3019, and Dual Locking Tape

SCIM & KEYBOARD:**NOTE:**

If Keyboard is to be replaced, also order **46-317475P2**, 2 inch x 4 inch adhesive backed velcro to attach to bottom of new keyboard.

KEYBOARDS:

- 5143798-7** Standard USB Keyboard – English
- 5143798-8** Standard USB Keyboard – French
- 5143798-9** Standard USB Keyboard – German
- 5143798-10** Standard USB Keyboard – Spanish
- 5143798-11** Standard USB Keyboard – Italian
- 5143798-12** Standard USB Keyboard – Portuguese
- 5143798-13** Standard USB Keyboard – Scandinavian
- 5183547-46** Standard USB Keyboard – Danish
- 5183547-47** Standard USB Keyboard – Dutch
- 5183547-48** Standard USB Keyboard – Norwegian

SCIM OVERLAYS:

- 5148389-2** MR SCIM – Black with English Overlay
- 5148389-9** MR SCIM – Black Coordinated Global Overlay Set

SERVICE KITS

CABLE FRU KIT's
5306523 Service Cable Hardware Kit
(Includes BNC components, studs, loopbacks, gender changers)

MRE HARDWARE

5408001
MR Elastography option kit
contains all MRE hardware



CABLE FRU'S**5331721, Scan Room Base SHORT Cable FRU Kit contains:**

Run M1302	Run M1303	Run M1304	Run M1310	Run M1311	Run M1312
Run M2025	Run M3030	Run M3031	Run M3300	Run M3301	Run M3302
Run M3303	Run M3304	Run M3306	Run M3307	Run M3308	Run M3309
Run M3310	Run M3311	Run M3312	Run M3313	Run M3315	Run M3316
Run M4005	13W6 Shunt Connector for 16 & 32 Channel Only				

5331721-2, Scan Room Base LONG Cable FRU Kit contains:

Run M1302	Run M1303	Run M1304	Run M1310	Run M1311	Run M1312
Run M2025	Run M3030	Run M3031	Run M3300	Run M3301	Run M3302
Run M3303	Run M3304	Run M3306	Run M3307	Run M3308	Run M3309
Run M3310	Run M3311	Run M3312	Run M3313	Run M3315	Run M3316
Run M4005	13W6 Shunt Connector for 16 & 32 Channel Only				

5343119, Fiber Optic Cable FRU Kit contains:

Run P2003	Run P2004
-----------	-----------

5343120, In-Room Display (IRD) Cable FRU Kit contains:

Run P2026	Run M3366
-----------	-----------

5352518, Display Cable FRU Kit contains:

Run P2026	Run M3366	Run M3314	Run M3363	Run M33349
-----------	-----------	-----------	-----------	------------

5169566	Run 3367, MAG-TDM-J1 to MAG-RX1, 44-pin HD sub-D 560mm
5169566-2	Run 3368, MAG-TDM-J2 to MAG-RX2, 44-pin HD sub-D 1360mm
5180907	Run 3350, MAG-SRI-J10 to MAG-BORE, Bore Temp, 3000mm
5329119-2	Run M1002, PEN-PNL J28 to RF-SW1 J121, 990mm
5329340-7	Run M1305, SPW-J122 to PED-IO-J1, 14306mm
5329340-8	Run M1305, SPW-J122 to PED-IO-J1, 18393mm
5329340-9	Run M1306, SPW-J120 to PED-IO-J2, 13965mm
5329340-10	Run M1306, SPW-J120 to PED-IO-J2, 18052mm
5329340-11	Run E1002, SPW-J128 to PGR-J18, 10218mm
5329340-12	Run E1002, SPW-J128 to PGR-J18, 16348mm
5330729-5	Run E1001, PGR-J19 to SPW-J106, 10327mm
5330729-6	Run E1001, PGR-J19 to SPW-J106, 16523mm
5330932-5	Run M1310, PEN-J72 to MAG-J9, 14000mm
5330932-26	Run M1312, PEN-J74 to MAG-HYB-J6, 17000mm
5330932-27	Run M1313, MAG-DDSPLIT-J20 to MAG-BORE-J6, 6250mm
5330932-28	Run M1314, MAG-DDSPLIT-J21 to MAG-BORE-J7, 6250mm
5344701	Run M1001, SPW-J106 to MAG-HYB-J1, 12394mm
5344701-2	Run M1001, SPW-J106 to MAG-HYB-J1, 16525mm

[illegible]

REVISION HISTORY

REV/DATE REASON FOR CHANGE

1 Feb 3, 2010 Initial release